

Day 2 — What Is CPI & How It Came About (25 Aug 2026)

1. What Is CPI?

CPI = Cloud Platform Integration

It is a **middleware tool**. Its core job: transfer data between systems.

Recap from Day 1: **SAP = Systems, Applications, and Products**.

2. SAP Modules — Functional vs Technical (Expanded)

Functional Modules	Technical Modules
SAP FICO (Finance)	SAP ABAP (UI)
SAP MM (Material Management)	SAP CPI
SAP SD (Sales & Distribution)	SAP BASIS
SAP HR (Human Resource)	SAP Security
	SAP PI/PO

There are around **40 modules** in the Functional category alone.

What the Technical/Admin-Focused Modules Actually Do

Module	What It Handles
SAP BASIS	Performs actions related to system administration — access, installation
SAP Security	Authorization, support, access control (in the cloud world, this maps to GRC — Governance, Risk & Compliance)
SAP PI/PO	A middleware tool that runs on-premise

3. On-Premise Modules → Their Cloud Equivalents

A key mapping from class — which older on-premise module corresponds to which modern cloud product:

On-Premise Module	Cloud Equivalent
SAP MM	SAP Ariba, SAP Concur, SAP Fiori, SAP SuccessFactors (SF)
SAP BASIS	BTP Admin
SAP Security	GRC
SAP PI/PO (middleware)	SAP BTP CPI / Integration Suite / CI

4. Worked Example — Honda & MRF (Expanded from Day 1)

Scenario: Honda is a manufacturing company. To accomplish the outcome for a bike, Honda needs tires from another company — MRF. So these two companies' systems need to contact each other.

This integration can happen in **four possible combinations**, depending on which systems each side is running:

Honda (SAP)	→ SD, FICO, PP, MM →	MRF (non-SAP)
Honda (SAP)	→	MRF (SAP)
Honda (non-SAP)	→	MRF (SAP)
Honda (non-SAP)	→	MRF (non-SAP)

This generalizes into the four possible data-flow directions CPI must be able to handle:

- SAP → SAP
- SAP → non-SAP
- non-SAP → SAP
- non-SAP → non-SAP

Important: the whole data itself is stored in **SAP ERP** — CPI's job is not to store data, but to move it.

5. Why Not Send Data Directly Between Systems?

This is a key conceptual question from class: **why don't we just send data directly to other systems, instead of using SAP CPI in between?**



The answer: **SAP ERP → ECC → S/4HANA is essentially a database.** When data is transferred directly from one system to another, this creates **security risks**.

How CPI solves this:

- CPI **reads the data** from the SAP ERP system
- CPI **only sends the requested/required data** to the third-party system — not the entire database
- CPI transfers data from one place to another **without storing the data itself** — it's a pass-through, not a data warehouse

Why this matters: If a system tried to connect directly to SAP ERP's database, it would potentially expose far more data than necessary, and create a much larger attack surface. CPI acts as a controlled gatekeeper — reading only what's needed and forwarding only what's requested, keeping the core system protected.

6. On-Premise vs Cloud

Example used in class — phone photos:

- **Phone gallery photos** (on-premise style): not able to access them without the phone itself
- **Cloud drive (Google Photos)** (cloud style): able to access from anywhere, as long as you have access to your Google account

This is the core distinction: **on-premise = tied to a specific physical location/device; cloud = accessible from anywhere with proper account access.**

On-Premise vs Cloud — Module Mapping

On-Premise	Cloud
SAP MM	SAP SF (SuccessFactors)
SAP PP	SAP C4C
SAP SD	SAP CDC
SAP ECC	SAP Ariba
SAP PI/PO (middleware)	SAP Concur
	SAP BTP & CPI (middleware)

7. Why Companies Moved From On-Premise to Cloud

Drawback of on-premise → reason companies moved to cloud: Accessibility & Upgradation

Real-World Analogy — Traveling from Hyderabad to Bangalore

Option	Model	What It Involves
1. Own Car	On-Premise	Buying the car, investment, fuel, driver, maintenance, repair, travel cost — everything is handled by the customer

2. High-Speed Train	Cloud	The vendor buys and owns the train — the customer only affords/pays for the travel cost , none of the ownership burden
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This captures the essence of cloud computing's appeal: you stop paying for and managing the underlying infrastructure yourself, and instead just pay for the service/usage — similar to how SAP BTP itself runs on top of rented hyperscaler infrastructure (AWS/Azure/GCP) rather than SAP owning physical datacenters for every customer.

8. Key Dates to Remember

Event	Year (from class notes)
SAP CPI started	2017
SAP stopping support (for ECC/on-premise)	2027

Verified/expanded during review:

- The **2027 date** your trainer mentioned is accurate — this is when **SAP ends mainstream maintenance for SAP ECC 6.0 (Enhancement Packages 6-8) and SAP Business Suite 7**. Note there's some nuance here: ECC 6.0 on EHP 0-5 already lost mainstream maintenance at the end of **2025**. Companies can pay for Extended Maintenance to continue support through **2030**, and select RISE with SAP customers can extend further to 2033 under specific conditions. This 2027 deadline is one of the single biggest drivers of the entire SAP job market right now — it's exactly why S/4HANA migration work (and by extension, CPI integration work supporting those migrations) is in such high demand.
- **SAP CPI's origin year** in your notes (2017) aligns with when SAP's cloud integration offering (then under the "SAP Cloud Platform Integration" name) became a distinct, actively developed product — though its technical roots trace back a bit earlier as part of the broader SAP HANA Cloud Platform/Cloud Platform Integration Services lineage. For interview purposes, 2017 is a reasonable answer if asked "when did CPI come about."